

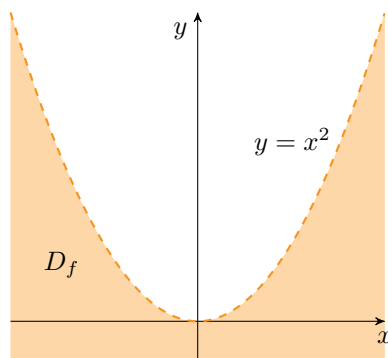
KUWAIT UNIVERSITY
Department of Mathematics

MATH 211: Calculus 3

Spring 2021–22 • First Midterm Exam • 20 April 2022

- Q1.** [10 pts] Find and sketch the domain of the function $f(x, y) = \frac{\sin(xy)}{\sqrt{x^2 - y}}$.
- Q2.** [15 pts] Use the two-path rule to show that $\lim_{(x,y) \rightarrow (0,0)} \frac{(x+2y)^2}{x^2 + y^2}$ does not exist.
- Q3.** [15 pts] Let $w = f(x, y)$ where $x = \sqrt{s^2 + t^2}$ and $y = st$. Given that f is differentiable with $f_x(5, 12) = -2$ and $f_y(5, 12) = 3$, find $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$ when $(s, t) = (3, 4)$.
- Q4.** [15 pts] Find an equation for the tangent plane to the surface $xz^2 + \frac{xy}{\sqrt{z}} + xy^2 = 21$ at the point $P(3, 2, 1)$.
- Q5.** [15 pts] Let $f(x, y) = \frac{e^{x(y-1)}}{x+2y}$.
- (a) Find the linearization of f at $P(2, 1)$.
 - (b) Find the directional derivative of f at $P(2, 1)$ in the direction from P toward the origin.
- Q6.** [15 pts] Find the four critical points of $f(x, y) = 3xy - x^2y - xy^2$ and classify them as local maximum, local minimum, or saddle points.
- Q7.** [15 pts] Use the Method of Lagrange Multipliers to find the point on the sphere $x^2 + y^2 + z^2 = 1$ that is closest to the point $P(2, -2, 1)$.

Q1. $D_f = \{(x, y) \mid x^2 - y > 0\}$ is the region below the parabola $y = x^2$, shaded below.



Q2. Restricted to the x -axis, the given limit becomes $L_1 = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$.

On the other hand, restricted to the y -axis it becomes $L_2 = \lim_{y \rightarrow 0} \frac{4y^2}{y^2} = 4$.

Since $L_1 \neq L_2$, the given limit does not exist.

Q3. When $(s, t) = (3, 4)$ we obtain $x = 5$ and $y = 12$. By the Chain Rule,

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} = \frac{s}{\sqrt{s^2 + t^2}} f_x + t f_y \Rightarrow \left. \frac{\partial w}{\partial s} \right|_{\substack{s=3 \\ t=4}} = \frac{3}{5} f_x(5, 12) + 4 f_y(5, 12) = \frac{54}{5},$$

and

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} = \frac{t}{\sqrt{s^2 + t^2}} f_x + s f_y \Rightarrow \left. \frac{\partial w}{\partial t} \right|_{\substack{s=3 \\ t=4}} = \frac{4}{5} f_x(5, 12) + 3 f_y(5, 12) = \frac{37}{5}.$$

Q4. Let $F(x, y, z) = xz^2 + \frac{xy}{\sqrt{z}} + xy^2$. Then

$$\nabla F = \left(z^2 + \frac{y}{\sqrt{z}} + y^2 \right) \mathbf{i} + \left(\frac{x}{\sqrt{z}} + 2xy \right) \mathbf{j} + \left(2xz - \frac{xy}{2z^{3/2}} \right) \mathbf{k} \Rightarrow \nabla F(3, 2, 1) = \langle 7, 15, 3 \rangle.$$

Hence an equation for the wanted tangent plane is

$$7(x - 3) + 15(y - 2) + 3(z - 1) = 0 \Rightarrow 7x + 15y + 3z = 54.$$

Q5. (a) We have $f(2, 1) = \frac{1}{4}$, $f_x = \frac{(y-1)(x+2y)e^{x(y-1)} - e^{x(y-1)}}{(x+2y)^2} \Rightarrow f_x(2, 1) = -\frac{1}{16}$, and

$$f_y = \frac{x(x+2y)e^{x(y-1)} - 2e^{x(y-1)}}{(x+2y)^2} \Rightarrow f_y(2, 1) = \frac{3}{8}. \text{ Thus the desired linearization is}$$

$$L(x, y) = \frac{1}{4} - \frac{1}{16}(x - 2) + \frac{3}{8}(y - 1) = -\frac{x}{16} + \frac{3y}{8}.$$

(b) By calculations of part (a), $\nabla f(2, 1) = -\frac{1}{16} \mathbf{i} + \frac{3}{8} \mathbf{j}$. On the other hand, the unit vector is

the direction from P to the origin is $\frac{\overrightarrow{PO}}{\|\overrightarrow{PO}\|} = \frac{1}{\sqrt{5}} \langle -2, -1 \rangle$. Hence

$$D_{\mathbf{u}} f(2, 1) = \nabla f(2, 1) \cdot \mathbf{u} = -\frac{1}{4\sqrt{5}} = -\frac{\sqrt{5}}{20}.$$

Q6. The partial derivatives $f_x = 3y - 2xy - y^2$ and $f_y = 3x - x^2 - 2xy$ are defined everywhere. Hence the critical points of f are solutions of the system $f_x = f_y = 0$. The equation $f_x = 0$ implies $y(3 - 2x - y) = 0$, and in turn, $y = 0$ or $3 - 2x - y = 0$. We solve the equation $f_y = 0$ in either case:

- $y = 0 \Rightarrow 3x - x^2 = 0 \Rightarrow x = 0, 3$. This gives the critical points $(0, 0)$ and $(3, 0)$.
- $3 - 2x - y = 0 \Rightarrow y = 3 - 2x \Rightarrow 3x - x^2 - 2x(3 - 2x) = 0 \Rightarrow -3x + 3x^2 = 0 \Rightarrow x = 0, 1$.
Since $y = 3 - 2x$, this gives the critical points $(0, 3)$ and $(1, 1)$.

To classify the critical points we use the discriminant

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 4xy - (3 - 2x - 2y)^2.$$

Since $D(0, 0) = D(3, 0) = D(0, 3) = -9 < 0$, the critical points $(0, 0)$, $(0, 3)$, and $(3, 0)$ correspond to saddle points of f . On the other hand, $D(1, 1) = 3 > 0$ and $f_{xx}(1, 1) = -2 < 0$, hence the critical point $(1, 1)$ corresponds to a local maximum of f .

Q7. Let d denote the distance between P and an arbitrary point (x, y, z) . We minimize $f(x, y, z) = d^2 = (x - 2)^2 + (y + 2)^2 + (z - 1)^2$ on the given sphere. Let $g(x, y, z) = x^2 + y^2 + z^2$. By the Method of Lagrange Multipliers, the desired minimum occurs at a point on the sphere $g = 1$ where $\nabla f = \lambda \nabla g$. That is, at a solution of the system

$$\begin{cases} 2(x - 2) = \lambda \cdot 2x \\ 2(y + 2) = \lambda \cdot 2y \\ 2(z - 1) = \lambda \cdot 2z \\ x^2 + y^2 + z^2 = 1 \end{cases} \Rightarrow \begin{cases} (\lambda - 1)x = -2 \\ (\lambda - 1)y = 2 \\ (\lambda - 1)z = -1 \\ x^2 + y^2 + z^2 = 1 \end{cases} \Rightarrow x = \frac{-2}{\lambda - 1}, y = \frac{2}{\lambda - 1}, z = \frac{-1}{\lambda - 1},$$

$$\Rightarrow x = 2z, y = -2z.$$

Together with the equation of the sphere these give $4z^2 + 4z^2 + z^2 = 1 \Rightarrow z = \pm \frac{1}{3}$. So the system has the solutions $\pm(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3})$. Now since $f(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}) = 4$ and $f(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}) = 16$, the point on the sphere closest to P is $(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3})$.